

WEEK:6-GIT-HOL

1.Git version:

```
apple@apples-MacBook-Pro-2 ~ % git --version  
git version 2.37.1 (Apple Git-137.1)
```

2.Verify Configuration:

```
[apple@apples-MacBook-Pro-2 ~ % git config --list  
credential.helper=osxkeychain  
init.defaultbranch=main  
user.name=sindhu24799  
user.email=sinduchevela@gmail.com
```

3.Create Project Folder:

```
apple@apples-MacBook-Pro-2 ~ % mkdir GitDemo
```

4.Initialize Git Repository:

```
[apple@apples-MacBook-Pro-2 ~ % cd GitDemo  
[apple@apples-MacBook-Pro-2 GitDemo % git init  
Initialized empty Git repository in /Users/apple/GitDemo/.git/  
[apple@apples-MacBook-Pro-2 GitDemo % ls -la  
total 0  
drwxr-xr-x  3 apple  staff   96 Jul 17 23:51 .  
drwxr-xr-x 120 apple  staff 3840 Jul 17 23:50 ..  
drwxr-xr-x  9 apple  staff  288 Jul 17 23:51 .git
```

5. Verify Hidden Files:

```
Initialized empty Git repository in /Users/apple/GitDemo/.git
[apple@apples-MacBook-Pro-2 GitDemo % ls -la
total 0
drwxr-xr-x  3 apple  staff   96 Jul 17 23:51 .
drwxr-x---+ 120 apple  staff 3840 Jul 17 23:50 ..
drwxr-xr-x  9 apple  staff  288 Jul 17 23:51 .git
```

6. Create File:

```
drwxr-xr-x  9 apple  staff  288 Jul 17 23:51 .git
[apple@apples-MacBook-Pro-2 GitDemo % touch welcome.txt
[apple@apples-MacBook-Pro-2 GitDemo % open welcome.txt
```

7. Verify File:

```
[apple@apples-MacBook-Pro-2 GitDemo % open welcome.txt
[apple@apples-MacBook-Pro-2 GitDemo % ls
welcome.txt
```

8. Read File:

```
[apple@apples-MacBook-Pro-2 GitDemo % cat welcome.txt
Git Hands-On file%
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        welcome.txt

nothing added to commit but untracked files present (use "git add" to track)
```

9. Check Status:

```
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        welcome.txt

nothing added to commit but untracked files present (use "git add" to track)
```

10.Add File:

```
apple@apples-MacBook-Pro-2 GitDemo % git add welcome.txt
apple@apples-MacBook-Pro-2 GitDemo % git add .
```

11.Check Status:

```
[apple@apples-MacBook-Pro-2 GitDemo % git status
[On branch main

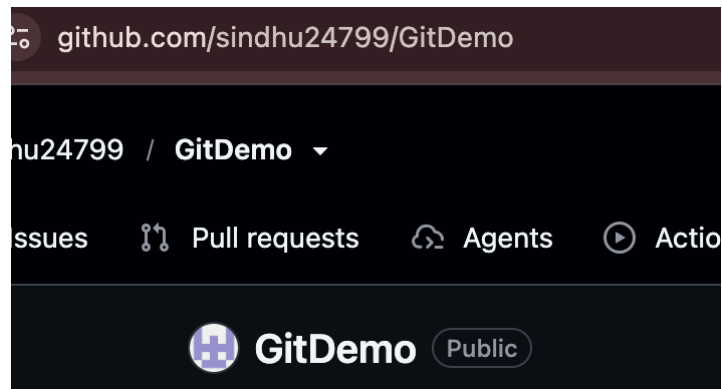
No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   welcome.txt
```

12.Commit:

```
apple@apples-MacBook-Pro-2 GitDemo % git commit -m"Added welcome file"
[main (root-commit) 61668fe] Added welcome file
 1 file changed, 1 insertion(+)
 create mode 100644 welcome.txt
```

13.Create GitHub Repository:



14.Connect Local Repository:

```
apple@apples-MacBook-Pro-2 GitDemo % git remote add origin https://github.com/sindhu24799/GitDemo.git
```

15. Verify Remote:

```
apple@apples-MacBook-Pro-2 GitDemo % git remote -v
origin https://github.com/sindhu24799/GitDemo.git (fetch)
origin https://github.com/sindhu24799/GitDemo.git (push)
apple@apples-MacBook-Pro-2 GitDemo % git branch -M main
```

16. Push:

```
apple@apples-MacBook-Pro-2 GitDemo % git branch -M main
apple@apples-MacBook-Pro-2 GitDemo % git push -u origin main
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/sindhu24799/GitDemo.git/'
apple@apples-MacBook-Pro-2 GitDemo % git pull origin main
```

HOL-2:

1. Create file:

```
apple@apples-MacBook-Pro-2 GitDemo % touch error.log
apple@apples-MacBook-Pro-2 GitDemo % mkdir log
apple@apples-MacBook-Pro-2 GitDemo % touch log/debug.txt
```

2. Check Git Status:

```
apple@apples-MacBook-Pro-2 GitDemo % git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    error.log
    log/

nothing added to commit but untracked files present (use "git add" to track)
```

3. Create .gitignore:

```
apple@apples-MacBook-Pro-2 GitDemo % touch .gitignore
apple@apples-MacBook-Pro-2 GitDemo % nano .gitignore
apple@apples-MacBook-Pro-2 GitDemo % open .gitignore
```

4. Verify:

```
[apple@apples-MacBook-Pro-2 GitDemo % git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore

nothing added to commit but untracked files present (use "git add" to track)
[apple@apples-MacBook-Pro-2 GitDemo % git add .gitignore
[apple@apples-MacBook-Pro-2 GitDemo % git commit -m "Added .gitignore file"
[main ccf0b5c] Added .gitignore file
1 file changed, 2 insertions(+)
create mode 100644 .gitignore
```

5. Add .gitignore:

```
[apple@apples-MacBook-Pro-2 GitDemo % git add .gitignore
[apple@apples-MacBook-Pro-2 GitDemo % git commit -m "Added .gitignore file"
[main ccf0b5c] Added .gitignore file
1 file changed, 2 insertions(+)
create mode 100644 .gitignore
[apple@apples-MacBook-Pro-2 GitDemo % git push
fatal: The current branch main has no upstream branch.
To push the current branch and set the remote as upstream, use

    git push --set-upstream origin main

To have this happen automatically for branches without a tracking
upstream, see 'push.autoSetupRemote' in 'git help config'.
```

Git-HOL 3: Branching and Merging:

Step 1. Create a New Branch:

```
nothing to commit, working tree clean
[apple@apples-MacBook-Pro-2 GitDemo % git branch GitNewBranch
```

Step 2: List All Branches:

```
[apple@apples-MacBook-Pro-2 GitDemo % git branch
* GitNewBranch
main
[apple@apples-MacBook-Pro-2 GitDemo % touch branch.txt
[apple@apples-MacBook-Pro-2 GitDemo % nano branch.txt
[apple@apples-MacBook-Pro-2 GitDemo % git status
On branch GitNewBranch
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        branch.txt

nothing added to commit but untracked files present (use "git add" to track)
```

Step 3: Switch to the New Branch:

```
[apple@apples-MacBook-Pro-2 GitDemo % git checkout GitNewBranch  
Switched to branch 'GitNewBranch'
```

Step 4: Create a New File:

```
main  
[apple@apples-MacBook-Pro-2 GitDemo % touch branch.txt  
[apple@apples-MacBook-Pro-2 GitDemo % nano branch.txt
```

Step 5: Check Status:

```
create mode 100644 branch.txt  
apple@apples-MacBook-Pro-2 GitDemo % git status  
On branch GitNewBranch  
nothing to commit, working tree clean
```

Step 6: Add the File:

```
nothing added to commit but untracked files present (use 'git add' to t  
[apple@apples-MacBook-Pro-2 GitDemo % git add branch.txt
```

Step 7: Commit the Changes:

```
1 file changed, 1 insertion(+)  
create mode 100644 branch.txt  
[apple@apples-MacBook-Pro-2 GitDemo % git commit -m "Added branch.txt in GitNewBranch"  
[GitNewBranch 43d06a7] Added branch.txt in GitNewBranch
```

Step 8: Verify Status:

```
apple@apples-MacBook-Pro-2 GitDemo % git status  
On branch GitNewBranch  
nothing to commit, working tree clean
```

Step 9: Switch Back to Main:

```
apple@apples-MacBook-Pro-2 GitDemo % git checkout main  
Switched to branch 'main'  
Your branch is up to date with 'origin/main'
```

Step 10: Compare Main and Branch:

```
apple@apples-MacBook-Pro-2 GitDemo % git diff main GitNewBranch
diff --git a/branch.txt b/branch.txt
new file mode 100644
index 0000000..542dab3
--- /dev/null
+++ b/branch.txt
@@ -0,0 +1 @@
+This file was created in GitNewBranch
```

Step 11: Merge the Branch:

```
[apple@apples-MacBook-Pro-2 GitDemo % git merge GitNewBranch
Updating ccf0b5c..43d06a7
Fast-forward
 branch.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 branch.txt
```

Step 12: View Commit History:

```
create mode 100644 branch.txt
[apple@apples-MacBook-Pro-2 GitDemo % git log --oneline --graph --decorate
* 43d06a7 (HEAD -> main, GitNewBranch) Added branch.txt in GitNewBranch
* ccf0b5c (origin/main) Added .gitignore file
* 61668fe Added welcome file
```

Step 13: Delete the Branch

```
[apple@apples-MacBook-Pro-2 GitDemo % git branch -d GitNewBranch
Deleted branch GitNewBranch (was 43d06a7)
```

Step 14: Verify

```
231. Command not found. Step
[apple@apples-MacBook-Pro-2 GitDemo % git branch
* main
```

Step 15: Push to GitHub

```
apple@apples-MacBook-Pro-2 GitDemo % git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 369 bytes | 369.00 KiB/s, done
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/sindhu24799/GitDemo.git
   cc5f0b5e..42d06a7  main -> main
```

Git-HOL 4: Merge Conflict Resolution

Step 1: Create a New Branch

```
apple@apples-MacBook-Pro-2 GitDemo % git checkout -b GitWork
Switched to a new branch 'GitWork'
```

Step 2: Create hello.xml:

```
apple@apples-MacBook-Pro-2 GitDemo % touch hello.xml
apple@apples-MacBook-Pro-2 GitDemo % nano hello.xml
```

Step 3: Commit the Changes:

```
apple@apples-MacBook-Pro-2 GitDemo % git add hello.xml
apple@apples-MacBook-Pro-2 GitDemo % git commit -m "Added hello.xml in GitWork"
[GitWork 7bfa0c5] Added hello.xml in GitWork
 1 file changed, 1 insertion(+)
 create mode 100644 hello.xml
```

Step 4: Switch to Main:

```
apple@apples-MacBook-Pro-2 GitDemo % git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```


Step 5: Modify the Same File Differently:

```
Your branch is up to date with 'origin/main'.
[apple@apples-MacBook-Pro-2 GitDemo % nano hello.xml
[apple@apples-MacBook-Pro-2 GitDemo % git add hello.xml
```

Step 6: Commit on Main:

```
[apple@apples-MacBook-Pro-2 GitDemo % git commit -m "Added hello.xml in main"
[main afc5458] Added hello.xml in main
 1 file changed, 1 insertion(+)
 create mode 100644 hello.xml
```

Step 7: View the Log:

```
[apple@apples-MacBook-Pro-2 GitDemo % git log --oneline --graph --decorate --all
* afc5458 (HEAD -> main) Added hello.xml in main
| * 7bfa0c5 (GitWork) Added hello.xml in GitWork
|/
* 43d06a7 (origin/main) Added branch.txt in GitNewBranch
* ccfc0b5c Added .gitignore file
* 61668fe Added welcome file
```

Step 8: Compare the Branches:

```
[apple@apples-MacBook-Pro-2 GitDemo % git diff main GitWork
diff --git a/hello.xml b/hello.xml
index 064b23b..0bf4c19 100644
--- a/hello.xml
+++ b/hello.xml
@@ -1,1 +1,1 @@
-<message>Hello from Main Branch</message>
```

Step 10: Merge:

```
<message>Hello from GitWork Branch</message>
[apple@apples-MacBook-Pro-2 GitDemo % git merge GitWork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed: fix conflicts and then commit the result
```

Step 11: List Branches:

```
[apple@apples-MacBook-Pro-2 GitDemo % git branch  
GitWork  
* main
```

Step 12: Delete the Branch:

```
[apple@apples-MacBook-Pro-2 GitDemo % git branch -d GitWork  
Deleted branch GitWork (was 7bfa0c5).
```

Step 13: View Final History:

```
Deleted branch GitWork (was 7bfa0c5).  
[apple@apples-MacBook-Pro-2 GitDemo % git log --oneline --graph --decorate  
* b569e01 (HEAD -> main) Updated .gitignore  
* 7201882 Resolved merge conflict  
|\  
| * 7bfa0c5 Added hello.xml in GitWork  
* | afc5458 Added hello.xml in main  
|/  
* 43d06a7 (origin/main) Added branch.txt in GitNewBranch  
* ccf0b5c Added .gitignore file  
* 61668fe Added welcome file
```

Git-HOL 5: Cleanup and Push to Remote Repository

Step 1: Check Working Tree:

```
[apple@apples-MacBook-Pro-2 GitDemo % git status
On branch main
Your branch is ahead of 'origin/main' by 4 commits.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
[apple@apples-MacBook-Pro-2 GitDemo % git branch -a
* main
remotes/origin/main
```

Step 2: List All Branches:

```
[apple@apples-MacBook-Pro-2 GitDemo % git branch -a
* main
remotes/origin/main
```

Step 3: Pull the Latest Changes

```
[apple@apples-MacBook-Pro-2 GitDemo % git pull origin main
From https://github.com/sindhu24799/GitDemo
 * branch                main                -> FETCH_HEAD
Already up to date.
```

Step 4: Push Local Changes

```
[apple@apples-MacBook-Pro-2 GitDemo % git push origin main
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 16 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (12/12), 1.17 KiB | 1.17 MiB/s, done.
Total 12 (delta 4), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (4/4), completed with 1 local object.
To https://github.com/sindhu24799/GitDemo.git
 43d06a7..b569e01  main -> main
```

Step 5: View Commit History

```
apple@apples-MacBook-Pro-2 GitDemo % git log --oneline --graph --decorate
* b569e01 (HEAD -> main, origin/main) Updated .gitignore
* 7201882 Resolved merge conflict
| \
|  * 7bfa0c5 Added hello.xml in GitWork
* |  afc5458 Added hello.xml in main
| /
* 43d06a7 Added branch.txt in GitNewBranch
* ccf0b5c Added .gitignore file
```